

Hemal Marepalli

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EDUCATION

Texas Tech University

B.S. in Computer Science, Concentration: Data Science, AI & Machine Learning

Lubbock, TX

Aug. 2023 - May 2027

- Minor in Mathematics
- GPA: 3.6/4.0

Relevant Coursework: Deep Learning, Introduction to Large Language Models, Data Structures & Algorithms, Computer Networks, Object-Oriented Programming

TECHNICAL SKILLS

Languages: Python, Java, C

Machine Learning / Computer Vision: PyTorch, torchvision, OpenCV, scikit-learn, Hugging Face Transformers

Deep Learning: CNNs, U-Net, Attention, Vision Transformers, Transfer Learning, LoRA

Data & Scientific Computing: NumPy, pandas, Matplotlib, Rasterio

Developer Tools: Git, GitHub, VS Code, Jupyter, Kaggle, Linux, CUDA

PROJECTS

MARIDA Marine Debris Segmentation | *PyTorch, U-Net, Attention, Rasterio*

- Built an 11-band Sentinel-2 semantic-segmentation pipeline and compared Plain U-Net, Attention U-Net, and Squeeze-and-Excitation variants across 11 ocean-surface classes.
- Addressed severe class imbalance (Marine Debris = 0.1955% of test pixels) using weighted cross-entropy, data augmentation, and mixed-precision training.
- Raised Marine Debris IoU from 0.1787 to 0.2428 (+35.9%) with spatial attention; Deep-SE achieved the best mean IoU (0.5913) and macro F1 (0.7075).

Multi-Cancer Image Classification | *PyTorch, Phikon, Qwen2-VL, LoRA*

- Developed three approaches for 26-class cancer and histopathology image classification: a custom CNN, frozen Phikon pathology backbone, and Qwen2-VL-2B with LoRA.
- Achieved 97.23% test accuracy and 0.971 macro F1 with Phikon transfer learning, an 8.92-point accuracy gain over the custom CNN (88.31%).
- Fine-tuned Qwen2-VL in 4-bit with LoRA while updating only 1.29% of model parameters, using Kaggle GPU training and mixed precision.

CIFAR-10 Image Classification | *Python, PyTorch, torchvision*

- Designed and trained a CNN with BatchNorm, ReLU, max pooling, dropout, data augmentation, early stopping, and learning-rate scheduling on 60,000 CIFAR-10 images.
- Reached 78.4% overall test accuracy, including 90.0% on automobiles and 86.6% on airplanes, and documented reproducible training, evaluation, and prediction outputs.